ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Nature of Matter: Elements, Compounds and Mixtures** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** element	That is the definition of an element.
2	**(c)** malleability	Malleable.
3	**(b)** a combination of two or more substances not chemically combined	Components keep properties.
4	**(c)** O_2	Oxygen contains atoms of only one kind.
5	**(b)** ductility	Ductile.
6	**(c)** in a fixed ratio by mass	Law of constant composition.
7	**(d)** CO_2	Carbon dioxide has two kinds of atoms chemically joined.
8	**(b)** sonority	Sonorous.
9	**(c)** mixture of gases	Mostly nitrogen and oxygen.
10	**(b)** Na	From the Latin natrium.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Pure substance made of only one kind of atom (1); e.g. iron, oxygen (1).	1+1
12	Element with properties between metals and non-metals (1); Silicon (1).	1+1
13	It is soft and leaves a mark (1); It conducts electricity (1).	1+1
14	Pure substance made of two or more elements chemically combined in a fixed ratio (1); e.g. water (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Lustre: gold (1); Malleability: aluminium foil (1); Conductivity: copper wire (1).	1+1+1
16	H and O are chemically combined (1); In a fixed ratio by mass (1); Water has properties different from H2 and O2 (1).	1+1+1
17	Smallest unit that exists independently of a substance (1); Elements: O_2, H_2 (1); Compounds: H_2O, CO_2 (1).	1+1+1
	OR: Generally dull (1); Brittle if solid (1); Poor conductors (except graphite) (1).	1+1+1
18	Mixture of metals or with non-metals (1); To improve strength or resistance to corrosion (1); Brass, bronze, steel (1).	1+1+1
	OR: Atoms of different elements have different properties (1); They have different sizes and masses (1); They are represented by different symbols (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Composition (1); Separation method (1); Properties (1); Examples (1); Fixed ratio or not (1).	1+1+1+1+1
	OR: H2: 2 hydrogen atoms (1); O2: 2 oxygen atoms (1); H2O: 2 H + 1 O (1); CO2: 1 C + 2 O (1); Molecule of an element vs compound (1).	1+1+1+1+1
20	Metals: lustre, malleability, ductility, conductivity, sonority (1); Non-metals: dull, brittle, poor conductors (1); Mercury is a liquid metal (1); Graphite conducts (1); Diamond is the hardest substance (1).	1+1+1+1+1
	OR: Evaporate to leave salt (1); Condense the vapour for pure water (1); Physical methods (1); Components keep properties (1); Both can be recovered (1).	1+1+1+1+1

Mistakes to avoid:

- Calling a molecule such as O_2 a compound; a compound needs *different* elements.
- Saying air or an alloy is a compound; they are mixtures.
- Assuming all metals are hard and solid; mercury is a liquid and sodium is soft.
- Writing wrong symbols (S for sodium, P for potassium); use Na and K.
- Using a physical method (filtration, magnet) to separate the elements of a compound.
- Writing that all non-metals are poor conductors; graphite is an exception.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Nature of Matter: Elements, Compounds and Mixtures** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** sulphur	Non-metals are usually brittle solids.
2	**(c)** salt solution	Uniform composition.
3	**(a)** K	From kalium.
4	**(d)** bromine	Bromine is a red-brown liquid.
5	**(c)** sand and water	Visible different parts.
6	**(b)** Fe	From ferrum.
7	**(d)** poor conductors of heat and electricity	Except graphite.
8	**(b)** copper and zinc	Brass.
9	**(d)** atom	Atoms are the building blocks.
10	**(a)** metalloids	e.g. silicon.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Short representation of an element (1); C and Ca (1).	1+1
12	(a) Hydrogen and oxygen (1); (b) Carbon and oxygen (1).	1+1
13	Metals: iron, copper (1); Non-metals: sulphur, oxygen (1).	1+1
14	Homogeneous: uniform, e.g. salt solution (1); Heterogeneous: not uniform, e.g. sand and water (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Mixture: no fixed composition; compound: fixed ratio (1); Mixture: separated by physical methods; compound: chemical methods (1); Mixture keeps properties; compound has new properties (1).	1+1+1
16	Oxygen for breathing (1); Carbon (as fuel and graphite) (1); Chlorine for water disinfection (1).	1+1+1
17	They are very unreactive (1); Helium, neon, argon (1); Used in signs and bulbs (1).	1+1+1
	OR: Heat a mixture of iron filings and sulphur (1); A black solid iron sulphide forms (1); It is not attracted by a magnet; new properties (1).	1+1+1
18	Iron for construction (1); Copper for wires (1); Aluminium for foils and utensils (1).	1+1+1
	OR: Lustre (1); Conductivity (1); Sonority and malleability (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Mixture: iron attracted by magnet (1); Mixture: components retain properties (1); After heating: black iron sulphide forms (1); Not magnetic (1); A new substance with new properties forms (1).	1+1+1+1+1
	OR: Iron: strong, buildings (1); Copper: conducts, wires (1); Aluminium: light, foils (1); Carbon: fuel, graphite (1); Oxygen: breathing (1).	1+1+1+1+1
20	Homogeneous: salt solution, air, brass (1); Heterogeneous: sand-water, oil-water, chalk powder (1); Filtration (1); Evaporation (1); Magnetic separation or separating funnel (1).	1+1+1+1+1
	OR: A compound has a fixed ratio of elements (1); Water: 2 H atoms and 1 O atom (1); Ratio by mass is 1 : 8 (1); Does not change with source (1); Mixtures vary (1).	1+1+1+1+1

Mistakes to avoid:

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- Saying air or an alloy is a compound; they are mixtures.
- Assuming all metals are hard and solid; mercury is a liquid and sodium is soft.
- Writing wrong symbols (S for sodium, P for potassium); use Na and K.
- Using a physical method (filtration, magnet) to separate the elements of a compound.

- Writing that all non-metals are poor conductors; graphite is an exception.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Nature of Matter: Elements, Compounds and Mixtures** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** physical methods	Filtration, evaporation, magnet...
2	**(c)** 2 hydrogen atoms and 1 oxygen atom	H_2O.
3	**(d)** silicon	Silicon is a metalloid.
4	**(a)** magnet	Iron is magnetic.
5	**(d)** graphite	Graphite is an allotrope of carbon.
6	**(b)** copper	Ductile and conducting.
7	**(b)** compound	New substance with new properties.
8	**(c)** mercury	Mercury is a liquid metal.
9	**(b)** lustrous and malleable	Shining and workable.
10	**(c)** distilled water	Distilled water is pure.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It is highly reactive (1); Reacts with moisture and oxygen in air (1).	1+1
12	Malleability: beaten into sheets (1); Ductility: drawn into wires (1).	1+1
13	Homogeneous mixture of metals or a metal with a non-metal (1); Brass or steel (1).	1+1
14	Metals conduct heat and handles would become hot (1); Plastic or wood handles are poor conductors (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Air: nitrogen, oxygen etc. (1); Sea water: water, salts (1); Soil: sand, clay, humus (1).	1+1+1
16	Use a magnet for iron (1); Add water, filter to remove sand (1); Evaporate to get salt (1).	1+1+1
17	Solids: iron, carbon (1); Liquids: mercury, bromine (1); Gases: hydrogen, oxygen (1).	1+1+1
	OR: $NaCl$: sodium, chlorine (1); H_2O: hydrogen, oxygen (1); CO_2: carbon, oxygen (1).	1+1+1
18	Sodium is a soft reactive metal; chlorine is a toxic gas (1); NaCl is a stable edible solid (1); New substance has new properties (1).	1+1+1
	OR: Sodium in kerosene (1); Phosphorus in water (1); To prevent reaction with air (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Metals: copper, aluminium (1); Non-metals: oxygen, carbon (1); Metalloids: silicon in chips (1); Choice depends on property (1); Alloys of metals (1).	1+1+1+1+1
	OR: Nitrogen about 78 per cent: inert (1); Oxygen about 21 per cent: supports burning (1); Argon and CO2: small amounts (1); Components keep properties (1); Proportions can vary (1).	1+1+1+1+1
20	Steel: iron and carbon, stronger (1); Brass: copper and zinc, hard and golden (1); Bronze: copper and tin, tough (1); Resist corrosion (1); Alloys are homogeneous mixtures (1).	1+1+1+1+1
	OR: Mixtures: physical separation (1); Magnet separates iron (1); Filtration separates sand (1); Compounds need chemical methods (1); Electrolysis of water (1).	1+1+1+1+1

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